

Steel Reinforcement Bar

Tensile Testing Guide

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1. Presentation Objectives

By the end of the presentation, participants should understand:

- Why tensile testing of reinforcing steel is required
- The mechanical properties determined by the test
- Applicable ASTM, ISO, EN and other standards
- Specimen preparation and measurement
- Correct installation in a universal testing machine (UTM)
- Tensile testing procedure
- Yield strength, tensile strength and elongation determination
- Stress–strain behavior of reinforcing steel
- Common testing errors
- Reporting requirements
- Equipment requirements and recommended laboratory configuration

2. What Is Rebar Tensile Testing?

Rebar tensile testing is a destructive mechanical test used to determine the strength and ductility of reinforcing steel bars.

A test specimen is subjected to a controlled **uniaxial tensile force** until fracture.

The test provides critical mechanical properties such as:

- Yield strength, f_y
- Tensile strength, f_u
- Yield-to-tensile ratio, f_u/f_y
- Elongation after fracture
- Uniform elongation, where required
- Stress–strain behavior
- Maximum tensile force
- Fracture characteristics

These properties are used to verify whether reinforcing steel satisfies the specified material standard and structural requirements.

3. Why Is Rebar Tensile Testing Important?

Reinforcing bars are fundamental structural components in:

- Reinforced concrete buildings
- Bridges
- Foundations
- Retaining walls
- Dams
- Tunnels
- Industrial structures
- Precast concrete
- Infrastructure projects

The tensile test verifies that the supplied reinforcement has adequate:

Strength + Ductility + Deformation Capacity

A bar can have adequate ultimate strength but still have insufficient ductility.

Therefore, **strength alone is not sufficient** for evaluating reinforcing steel.

4. Fundamental Tensile-Test Concept

The basic tensile test can be represented as:

Specimen → Tensile Load → Deformation → Yield → Ultimate Load → Necking → Fracture

The applied tensile force is measured continuously while the specimen elongates.

The engineering stress is:

$$\sigma = \frac{F}{A_0}$$

where:

- σ = engineering stress
- F = applied tensile force
- A_0 = original cross-sectional area

Engineering strain is:

$$\varepsilon = \frac{\Delta L}{L_0}$$

where:

- ε = engineering strain
- ΔL = change in gauge length
- L_0 = original gauge length

5. Typical Stress–Strain Curve

A typical mild/ductile reinforcing steel stress–strain curve contains:

- **Elastic region**
- **Yield point / yield region**
- **Plastic deformation**
- **Strain hardening**
- **Ultimate tensile strength**
- **Necking**
- **Fracture**

Important observation

The area under the stress–strain curve represents the material's ability to absorb mechanical energy.

A ductile reinforcing steel normally exhibits substantial plastic deformation before fracture.

6. Main Mechanical Properties

Yield Strength

- Yield strength is the stress corresponding to the onset of significant plastic deformation, according to the applicable standard.
- It is generally reported as:
- $f_y = \frac{F_y}{A_0}$
- where F_y is the yield force.
- For some reinforcing steels, a distinct yield point is visible.
- For others, the yield strength is determined using a **proof-stress method**, commonly the **0.2% proof stress**.

7. Tensile Strength

- The tensile strength or ultimate tensile strength is:
- $f_u = \frac{F_m}{A_0}$
- where:
- F_m = maximum tensile force
- A_0 = original cross-sectional area
- It represents the maximum engineering stress reached during the tensile test.

8. Yield-to-Tensile Ratio

- An important parameter for reinforcing steel is:
- $\frac{f_u}{f_y}$
- A higher ratio generally indicates greater reserve strength between yielding and ultimate tensile failure.
- This ratio is particularly important for seismic and structural design requirements where adequate ductility and plastic deformation capacity are required.

9. Elongation

- Elongation measures the permanent extension of the specimen after fracture.
- A common expression is:
- $A = \frac{L_u - L_0}{L_0} \times 100$
- where:
- L_0 = original gauge length
- L_u = gauge length after fracture and fitting the fractured pieces together
- Elongation is normally reported as a percentage.

10. Uniform Elongation

- Some standards require determination of **uniform elongation**, which represents deformation occurring before localized necking.
- This parameter can be especially useful for evaluating ductility and seismic performance.
- The exact measurement and acceptance requirements must be taken from the applicable product standard.

11. Rebar Geometry

Unlike ordinary round tensile specimens, reinforcing bars have:

- Ribs
- Deformations
- Variable surface geometry
- Nominal diameter
- Nominal cross-sectional area

Therefore, determining the correct specimen dimensions and cross-sectional area is essential.

For many rebar standards, the **nominal cross-sectional area** is used for strength calculations rather than simply calculating the area from an arbitrary measured diameter.

12. Common Rebar Sizes

Typical reinforcing bar sizes include:

- 8 mm
- 10 mm
- 12 mm
- 16 mm
- 20 mm
- 25 mm
- 32 mm
- 40 mm

Larger diameters may require substantially higher-capacity testing machines and suitable grips.

13. Test Standards

- **ASTM A370**
Standard Test Methods and Definitions for Mechanical Testing of Steel Products.
- **ASTM A615/A615M**
Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.
- **EN ISO 15630-1**
Steel for the reinforcement and prestressing of concrete — Test methods for reinforcing bars, wire rod and wire.

	ISO	ASTM
Rebar Product Standard	6935-2	A615
Rebar Testing Standard	15630-1	A370
Metals Tensile Test Standard	6892-1	E8

14. Test Equipment

A professional rebar tensile testing system normally consists of:

- **1. Universal Testing Machine**
- **2. Rebar Grips**
- **3. Load Cell**
- **4. Extensometer**
- **5. Displacement Measurement System**
- **6. Control and Data Acquisition System**

15. Specimen Selection

The specimen should be representative of the material being evaluated.

Record:

- Manufacturer
- Heat number
- Batch/lot number
- Bar designation
- Nominal diameter
- Grade
- Test direction
- Sampling location
- Sample identification

Avoid specimens with obvious:

- Severe corrosion
- Mechanical damage
- Deep scratches
- Unauthorized machining
- Excessive bending
- unless the purpose of the test specifically requires evaluating those conditions.

16. Specimen Preparation

The test specimen is generally cut to the required length.

The cutting operation should avoid excessive heating or mechanical damage.

Recommended practice:

- Use an appropriate cutting method
- Avoid localized overheating
- Avoid bending the specimen
- Keep the original rib geometry intact where required
- Mark the gauge length
- Record specimen identification

17. Gauge Length

- The gauge length depends on the applicable standard.
- Common gauge-length conventions may be expressed as a function of nominal bar diameter or cross-sectional area.
- For example, certain European tensile-testing procedures use gauge-length relationships based on the specimen geometry.
- **Do not assume one gauge length applies to every standard.**
- The test software should allow the operator to enter the required gauge length according to the selected standard.

18. Initial Measurements

Before testing, record:

- Nominal diameter
- Nominal area
- Specimen length
- Gauge length
- Bar identification
- Grade
- Manufacturer
- Batch/heat number

For research or investigation purposes, additional measurements may include:

- Actual mass per unit length
- Rib height
- Rib spacing
- Actual dimensions
- Surface condition

19. Installing the Rebar

The specimen must be installed **coaxially** with the machine loading axis.

Incorrect alignment can cause:

- Bending stresses
- Uneven grip loading
- Premature fracture
- Incorrect yield measurement
- Reduced measured tensile strength

Key principle

The tensile force must pass through the longitudinal axis of the bar.

20. Grip Design

Rebar grips are one of the most important components of the test system.

A suitable grip must withstand the maximum tensile force without slippage.

Common approaches include:

- Wedge grips
- Hydraulic wedge grips
- Serrated grips
- Specialized rebar grips
- Threaded/end fixtures for specially prepared specimens

For ordinary deformed reinforcing bars, the gripping system should be selected specifically for the bar diameter, geometry and required test force.

21. Extensometer

An extensometer measures strain directly on the specimen.

It can be used to determine:

- Yield strain
- Elastic modulus, where required
- Proof stress
- Uniform elongation
- Stress–strain curve

The extensometer must have:

- Suitable gauge length
- Adequate travel
- Appropriate accuracy
- Secure attachment
- Compatibility with the specimen geometry

22. Testing Procedure

- **Step 1 — Inspect the machine**
 - Check:
 - Hydraulic system
 - Load cell
 - Grips
 - Emergency stop
 - Control system
 - Extensometer
 - Calibration status
- **Step 2 — Identify the specimen**
 - Enter:
 - Sample ID
 - Grade
 - Diameter
 - Standard
 - Batch/heat
- **Step 3 — Install the specimen**
 - Ensure proper axial alignment.
- **Step 4 — Install the extensometer**
 - Attach it according to the applicable standard.

22. Testing Procedure

- **Step 5 — Zero the system**
- Zero:
- Force
- Extension
- Displacement
- Strain
- **Step 6 — Start the test**
- Apply tensile loading at the specified rate.
- **Step 7 — Record the complete curve**
- Capture:
- **Force vs. extension**
or
- **Stress vs. strain**
- **Step 8 — Remove the extensometer**
- If required, remove it before fracture to protect the instrument.
- **Step 9 — Continue loading**
- Continue until fracture.
- **Step 10 — Examine the fracture**
- Record fracture location and appearance.

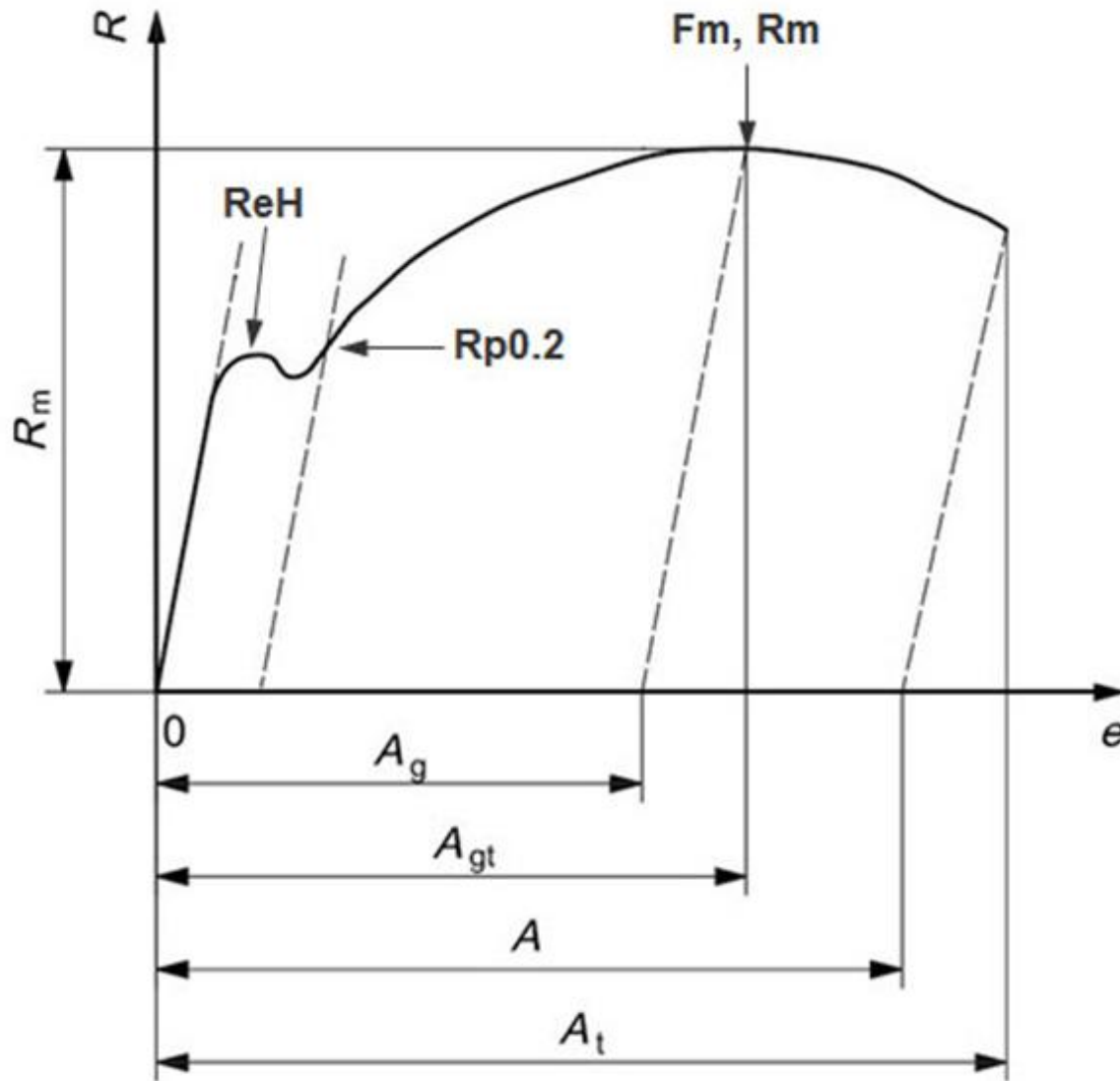
23. Loading Rate

ISO 6892-1:2009

ASTM A370-14

ELASTIC REGION (crosshead separation rate)	0.00025/sec * Lc 3 mm/min	0.0625 in/min * GL 1.6 mm/min
ELASTIC REGION (strain rate)	0.00025/sec	Not Specified
ELASTIC REGION (stress rate)	6 – 60 MPa/sec 2-20 mp/s	10-100 ksi/min 1-12 mp/s
YIELDING (crosshead separation rate)	0.00025/sec * Lc	0.0625 in/min * GL
PLASTIC REGION (crosshead separation rate)	0.0067/sec * Lc	0.5 in/min * GL

24. Typical Test Curve



ISO

ASTM

Yield Point (distinct)	Upper Yield Strength (ReH)	Yield Point (Drop of Beam or Halt of Pointer)
Yield Strength (Offset Method)	0.2% Proof Strength, non-proportional elongation (Rp 0.2)	Yield Strength (0.2% Offset)
Maximum Stress	Tensile Strength (Rm)	Tensile Strength
Ratio of Tensile Strength/Yield Strength	Rm/ReH	Not Required
Strain at Maximum Force	% Total Elongation at Maximum Force (Agt)	Not Required
Elongation after Fracture	% Elongation After Fracture (A or A ₅)	Percent Elongation

25. Example Calculation

- Assume:
- Nominal area = 201 mm²
- Yield force = 105 kN
- Maximum force = 135 kN
- Yield strength:
 - $f_y = \frac{105000}{201}$
 - $f_y \approx 522 \text{ MPa}$
- Tensile strength:
 - $f_u = \frac{135000}{201}$
 - $f_u \approx 672 \text{ MPa}$
- Yield-to-tensile ratio:
 - $\frac{f_u}{f_y} = \frac{672}{522} \approx 1.29$
- The actual acceptance of these values depends on the specified steel grade and applicable product standard.

26. Common Testing Errors

- **Error 1 — Incorrect alignment**
 - Causes additional bending.
 - **Solution:** Carefully align the specimen with the machine axis.
- **Error 2 — Grip slippage**
 - Causes incorrect extension and potentially invalid results.
 - **Solution:** Use appropriate rebar grips and sufficient gripping force.
- **Error 3 — Incorrect cross-sectional area**
 - Can significantly change calculated stress.
 - **Solution:** Use the area definition required by the applicable standard.

26. Common Testing Errors

- **Error 4 — Incorrect loading rate**
 - Can influence measured mechanical properties.
 - **Solution:** Use standard-compliant test control.
- **Error 5 — Incorrect gauge length**
 - Produces incorrect elongation results.
 - **Solution:** Follow the specified standard exactly.
- **Error 6 — Extensometer problems**
 - Incorrect attachment can distort strain measurements.
 - **Solution:** Verify gauge length, attachment and calibration.

27. Safety

Rebar tensile testing is a **high-energy destructive test**.

Potential hazards include:

- Sudden specimen fracture
- Flying steel fragments
- Grip failure
- Hydraulic pressure
- Machine movement
- High tensile forces

Safety requirements

- Use machine guarding.
- Keep personnel outside the danger zone.
- Never stand directly in line with the specimen.
- Verify grip engagement.
- Inspect hydraulic components.
- Use emergency-stop functionality.
- Do not touch the specimen during loading.
- Follow the manufacturer's safety instructions.

Thank you for your attention

For more questions, please:

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